

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

**Product Name** Palladium, 5% on calcium carbonate, lead poisoned, A305060-5

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>H36186</b>                                                                               |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <u>ANZinfo@thermofisher.com</u>                                                             |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

|                                   |             |
|-----------------------------------|-------------|
| Serious Eye Damage/Eye Irritation | Category 2  |
| Germ Cell Mutagenicity            | Category 2  |
| Carcinogenicity                   | Category 2  |
| Reproductive Toxicity             | Category 1A |
| /; Effects on or via lactation    |             |

#### Environmental hazards

|                          |            |
|--------------------------|------------|
| Chronic aquatic toxicity | Category 2 |
|--------------------------|------------|

### Label Elements



Signal Word

Danger

#### Hazard Statements

H360 - May damage fertility or the unborn child  
H411 - Toxic to aquatic life with long lasting effects  
H319 - Causes serious eye irritation  
H341 - Suspected of causing genetic defects if inhaled  
H351 - Suspected of causing cancer  
H362 - May cause harm to breast-fed children

#### Precautionary Statements

##### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P273 - Avoid release to the environment

##### Response

P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P391 - Collect spillage

##### Storage

P405 - Store locked up

##### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

Other hazards which do not result in classification

## Section 3 - Composition and Information on Ingredients

| Component         | CAS No    | Weight % |
|-------------------|-----------|----------|
| Calcium carbonate | 471-34-1  | 92.0     |
| Palladium         | 7440-05-3 | 5.0      |
| Lead              | 7439-92-1 | 3.0      |

## Section 4 - First Aid Measures

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### New Zealand Emergency Tel.

CHEMTREC®  
09 980 6780 or +64 9 980 6780

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                            |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                                                     |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                           |

## Section 5 - Fire Fighting Measures

**Suitable Extinguishing Media**  
Not combustible.

**Extinguishing media which must not be used for safety reasons**  
No information available.

**Specific Hazards Arising from the Chemical**  
Thermal decomposition can lead to release of irritating gases and vapors.

**Hazardous Combustion Products**  
Calcium oxides, lead oxides, Palladium oxide.

**Special protective equipment and precautions for fire fighters**  
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

**Emergency procedures**  
Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**  
Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.

**Methods for Containment and Clean Up**  
Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

**Precautions to prevent secondary hazards**  
Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**  
Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

**Advice on safe handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

#### Conditions for Safe Storage, Including any Incompatibilities

##### Storage Conditions

Keep container tightly closed in a dry and well-ventilated place.

##### Incompatible Materials

Organic materials.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

#### Control parameters

##### Exposure limits

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

| Component         | New Zealand WEL             | Australia                   | ACGIH TLV                   | The United Kingdom                                                      |
|-------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------------------------------------------------------|
| Calcium carbonate | TWA: 10 mg/m <sup>3</sup>   | TWA: 10 mg/m <sup>3</sup>   |                             |                                                                         |
| Lead              | TWA: 0.05 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup> | STEL: 0.45 mg/m <sup>3</sup> 15 min<br>TWA: 0.15 mg/m <sup>3</sup> 8 hr |

##### Biological limit values

**NZ** - Substances assigned Biological Exposure Indices in the New Zealand Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**ACGIH** - American Conference of Governmental Industrial Hygienists (ACGIH) TLVs® and BEIs®- Threshold Limit Values for Chemical Substances and Physical Agents & Biological Exposure Indices. 2022 Edition

| Component | New Zealand                                                                                                                                                       | Australia | ACGIH - Biological Exposure Indices                                  | United Kingdom |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------|----------------|
| Lead      | 3 µg/dL (blood) not critical (Lead)<br>0.14 µmol/L (blood) not critical (Lead)<br>0.48 µmol/L (blood) not critical (Lead)<br>10 µg/dL (blood) not critical (Lead) |           | 200 µg/L<br>Medium: blood<br>Time: not critical<br>Determinant: Lead |                |

#### Appropriate engineering controls

##### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Individual protection measures, such as personal protective equipment

## Eye Protection

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

## Hand Protection

Protective gloves

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber, Nitrile rubber, Neoprene, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Skin and body protection

Long sleeved clothing

## Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

### Recommended Filter type: Recommended half mask:-

Particulates filter conforming to EN 143 (or AUS/NZ equivalent)  
Particle filtering: EN149:2001 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

# Section 9 - Physical and Chemical Properties

## Information on basic physical and chemical properties

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Physical State                          | Solid                    |                                   |
| Appearance                              |                          |                                   |
| Odor                                    | Odorless                 |                                   |
| Odor Threshold                          | No data available        |                                   |
| pH                                      | No information available |                                   |
| Melting Point/Range                     | No data available        |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | No information available |                                   |
| Flammability (liquid)                   | Not applicable           | Solid                             |
| Flammability (solid,gas)                | No information available |                                   |
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | No information available | Method - No information available |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| Viscosity                               | Not applicable           | Solid                             |
| Water Solubility                        | Insoluble in water       |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Vapor Pressure                          | 23 hPa @ 20 °C           |                                   |
| Density / Specific Gravity              | No data available        |                                   |
| Bulk Density                            | No data available        |                                   |
| Vapor Density                           | Not applicable           | Solid                             |
| Particle characteristics                | No data available        |                                   |

Other information

Evaporation Rate Not applicable - Solid

## Section 10 - Stability and Reactivity

Reactivity None known, based on information available

Stability Stable under normal conditions.

Sensitivity to Mechanical Impact No information available

Sensitivity to Static Discharge No information available

Hazardous Polymerization No information available.

Hazardous Reactions None under normal processing.

Conditions to Avoid Heat, flames and sparks.

Incompatible Materials Organic materials.

Hazardous Decomposition Products Calcium oxides. lead oxides. Palladium oxide.

## Section 11 - Toxicological Information

Acute Effects

Information on likely routes of exposure

**Product Information**

Inhalation Not an expected route of exposure.  
Eyes Avoid contact with eyes.  
Skin Avoid contact with skin.  
Ingestion May be harmful if swallowed.

Numerical measures of toxicity

**(a) acute toxicity;**

Oral Based on available data, the classification criteria are not met  
Dermal No data available  
Inhalation No data available

Toxicology data for the components

| Component         | LD50 Oral                 | LD50 Dermal                                | LC50 Inhalation                           |
|-------------------|---------------------------|--------------------------------------------|-------------------------------------------|
| Calcium carbonate | LD50 = 6450 mg/kg ( Rat ) | > 2000 mg/kg (Rat)<br>(OECD Guideline 402) | > 3 mg/L 4h (Rat)<br>(OECD Guideline 403) |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

**(d) respiratory or skin sensitization;**

Respiratory No data available  
Skin No data available

| Component                              | Test method                                       | Test species | Study result    |
|----------------------------------------|---------------------------------------------------|--------------|-----------------|
| Calcium carbonate<br>471-34-1 ( 92.0 ) | OECD Test Guideline 429 Local<br>Lymph Node Assay | mouse        | non-sensitising |

(e) germ cell mutagenicity; No data available

| Component                              | Test method                                                | Test species | Study result |
|----------------------------------------|------------------------------------------------------------|--------------|--------------|
| Calcium carbonate<br>471-34-1 ( 92.0 ) | Bacterial Reverse Mutation Test<br>OECD Test Guideline 471 | in vitro     | negative     |

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | New Zealand             | Australia | New South<br>Wales | Western<br>Australia | IARC     | EU | UK | Germany |
|-----------|-------------------------|-----------|--------------------|----------------------|----------|----|----|---------|
| Lead      | Suspected<br>carcinogen |           |                    |                      | Group 2A |    |    |         |

(g) reproductive toxicity; Category 1A

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and delayed  
No information available.

## Section 12 - Ecological Information

### Ecotoxicity

#### Aquatic ecotoxicity

The product contains following substances which are hazardous for the environment. Contains a substance which is: Very toxic to aquatic organisms. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component         | Freshwater Fish                                                                                                                                                                            | Water Flea                            | Freshwater Algae | Microtox |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------|----------|
| Calcium carbonate | LC50 > 56 g/L/96h                                                                                                                                                                          |                                       |                  |          |
| Lead              | LC50: = 1.32 mg/L, 96h<br>static (Oncorhynchus<br>mykiss)<br>LC50: = 1.17 mg/L, 96h<br>flow-through<br>(Oncorhynchus mykiss)<br>LC50: = 0.44 mg/L, 96h<br>semi-static (Cyprinus<br>carpio) | EC50: = 600 µg/L, 48h<br>(water flea) |                  |          |

Terrestrial ecotoxicity There is no data for this product

Persistence and Degradability Product contains heavy metals. Discharge into the environment must be avoided. Special

|                                                                                                                   |                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                   | pre-treatment is necessary                                                                                                                                                                                    |
| <b>Persistence</b>                                                                                                | Insoluble in water, May persist.                                                                                                                                                                              |
| <b>Degradability</b><br><b>Degradation in sewage treatment plant</b>                                              | Not relevant for inorganic substances.<br>Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.                                                     |
| <b>Bioaccumulative Potential</b>                                                                                  | May have some potential to bioaccumulate Product has a high potential to bioconcentrate                                                                                                                       |
| <b>Mobility</b>                                                                                                   | Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility.                                                                                                    |
| <b><u>Other adverse effects</u></b>                                                                               |                                                                                                                                                                                                               |
| <b>Endocrine Disruptor Information</b><br><b>Persistent Organic Pollutant</b><br><b>Ozone Depletion Potential</b> | This product does not contain any known or suspected endocrine disruptors<br>This product does not contain any known or suspected substance<br>This product does not contain any known or suspected substance |

## Section 13 - Disposal Considerations

### Waste treatment methods

|                                            |                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.       |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                                  |
| <b>Other Information</b>                   | Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment. |

## Section 14 - Transport Information

|                                                                                 |                                                                                                          |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b><u>NZS 5433:2020</u></b>                                                     | Not regulated                                                                                            |
| <b><u>IATA</u></b>                                                              | Not regulated                                                                                            |
| <b><u>IMDG/IMO</u></b>                                                          | Not regulated                                                                                            |
| <b>Environmental hazards</b>                                                    | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
| <b>Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code</b> | Not applicable, packaged goods                                                                           |
| <b>Special Precautions</b>                                                      | No special precautions required. Please refer to the applicable dangerous goods                          |



regulations for additional information.

Additional information

None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

| Component | New Zealand          |
|-----------|----------------------|
| Lead      | Suspected carcinogen |

### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

#### Authorisation/Restrictions according to EU REACH

| Component         | REACH (1907/2006) - Annex XIV -<br>Substances Subject to<br>Authorization | REACH (1907/2006) - Annex XVII -<br>Restrictions on Certain Dangerous<br>Substances                                                                                                                                                                                                  | REACH Regulation (EC<br>1907/2006) article 59 - Candidate<br>List of Substances of Very High<br>Concern (SVHC) |
|-------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Calcium carbonate | -                                                                         | Use restricted. See item 75.<br>(see link for restriction details)                                                                                                                                                                                                                   | -                                                                                                              |
| Lead              | -                                                                         | Use restricted. See item 72.<br>(see link for restriction details)<br>Use restricted. See item 30.<br>(see link for restriction details)<br>Use restricted. See item 63.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | SVHC Candidate list - 231-100-4 -<br>Toxic for reproduction (Article 57c)                                      |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/authorisation-list>  
<https://echa.europa.eu/substances-restricted-under-reach>  
<https://echa.europa.eu/candidate-list-table>

### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component         | CAS No    | NZIoC | AICS | EINECS | ELINCS | NLP | KECL     | IECSC | TCSI |
|-------------------|-----------|-------|------|--------|--------|-----|----------|-------|------|
| Calcium carbonate | 471-34-1  | X     | X    | -      | -      | -   | KE-04487 | X     | X    |
| Palladium         | 7440-05-3 | X     | X    | -      | -      | -   | KE-27744 | X     | X    |
| Lead              | 7439-92-1 | X     | X    | -      | -      | -   | KE-21887 | X     | X    |

| Component         | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|-------------------|-----------|------|-----------------------------------------------|-----|------|-------|------|------|
| Calcium carbonate | 471-34-1  | X    | ACTIVE                                        | X   | -    | X     | X    | X    |
| Palladium         | 7440-05-3 | X    | ACTIVE                                        | X   | -    | X     | -    | X    |
| Lead              | 7439-92-1 | X    | ACTIVE                                        | X   | -    | X     | -    | X    |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations

### Legend

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**NZS 5433:2020** - Transport of Dangerous Goods on Land

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**PNEC** - Predicted No Effect Concentration

**OECD** - Organisation for Economic Co-operation and Development

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

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First aid for chemical exposure, including the use of eye wash and safety showers.

|                  |                |
|------------------|----------------|
| Revision Date    | 22-Mar-2023    |
| Revision Summary | Not applicable |

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet